

KAYALVIZHI SHANMUGAM

Enterprise Java Solution Architect | AWS & Azure Cloud | Microservices & API Strategy

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Professional Summary

Enterprise Java Solution Architect with 17+ years of experience defining and delivering scalable, secure, and high-performance distributed systems across AWS and Azure. Specialized in designing enterprise-grade API platforms, microservices ecosystems, and event-driven architectures handling high-volume transactional workloads with strict SLA and compliance requirements.

Proven expertise in leading large-scale cloud modernization initiatives, establishing architecture governance, and driving platform standardization across multi-team environments. Strong track record in optimizing cost, performance, and resilience through strategic architectural decisions, including service mesh adoption, zero-trust security models, and observability frameworks.

Currently focused on integrating AI-driven capabilities (Agentic AI, RAG, LLMs) into enterprise platforms for intelligent automation and self-healing systems.

Availability Immediate Joiner | Open to Full-Time / Contract

Core Expertise

Enterprise Architecture and Solution Design for cloud-native, microservices, and event-driven systems across AWS and Azure, ensuring scalability, resilience, and performance.

Strong experience in architecture governance, including design reviews, standards definition, API strategy, and implementation of secure, compliant, and reusable enterprise platforms

Architecture Leadership & Governance

Defined enterprise-wide microservices and API architecture standards across AWS and Azure

Led architecture reviews, design governance, and solution approvals for distributed platforms

Established cloud-native design principles including resilience, scalability, and observability

Standardized security architecture (OAuth2, mTLS, Zero Trust) across enterprise services

Designed reusable reference architectures accelerating onboarding and delivery consistency

Technical Skills

Architecture	Cloud Architecture, Platform Engineering, Microservices, Event-Driven Architecture, Distributed Systems, Service Mesh (Istio), API Strategy
AI & GenAI	Agentic AI, LLM Integration, RAG/GraphRAG, Prompt Engineering, Vector Databases (FAISS, OpenSearch), AWS Bedrock, LangChain4j
Cloud	AWS (EKS, Lambda, CloudFormation, SQS/SNS), Azure (ACR, AKS, APIM, ADF, Monitor)
DevOps	Terraform, Docker, Kubernetes, Jenkins, GitHub Actions, Azure DevOps
Programming	Java (8/11/17/21), Python, Node.js, TypeScript, Reactive Programming, React.js
Frameworks	Spring Boot, Spring Cloud, Spring WebFlux, Hibernate, JUnit, Mockito
Integration	Apache Camel (EIP), REST, GraphQL, gRPC, API Gateway, Kafka, RabbitMQ, Event Hub
Orchestration	Camunda (BPMN 2.0), IBM BPM, Azure Data Factory (ADF), Workflow Automation
Observability	Prometheus, Grafana, ELK, Datadog, Splunk, OpenTelemetry
Security	OAuth2, JWT, mTLS, SAML, Zero Trust, WAF (Custom Java-based)
Database	RDBMS (MySQL, PostgreSQL, Oracle, DB2), NoSQL (MongoDB, DynamoDB, Redis, Neo4j)

Professional Experience

- Jan 2025–Pres. **Senior Solution Architect, Independent Consulting (Contract)**
- Architected enterprise-grade Agentic AI platforms using Java 17 and LangChain4j, leveraging LLMs and GraphRAG (Neo4j) to enable autonomous workflows for real-time log analysis and root cause detection.
 - Designed reactive microservices using Spring WebFlux integrated with vector databases (FAISS/OpenSearch), reducing API response latency by 40%.
 - Engineered self-healing cloud-native platforms by integrating Agentic AI with Istio service mesh telemetry, enabling automated closed-loop remediation and reducing production incidents by 35%.
- Oct 2024–Dec 2024 **Senior Technical Lead (End Client: Microsoft), Cognizioni (Deputed to Capgemini)**
- Led Azure cloud modernization initiatives defining AKS-based architecture for containerized workloads.
 - Resolved critical scalability bottlenecks improving system availability to 99.9% through optimized traffic management.
 - Established a robust observability architecture using Azure Monitor and Splunk, cutting incident resolution time significantly.
- Sep 2020–Jun 2024 **Technical Lead III (Client: MedImpact Healthcare), Photon**
- Designed a High-Availability Multi-Cloud Platform across AWS and Azure using Terraform (IaC) and Kubernetes (EKS/AKS); implemented automated disaster recovery and fault-tolerant patterns that ensured 99.99% uptime and 100% adherence to SOC2/GDPR enterprise compliance standards.
 - Architected a Zero Trust Security Framework by designing a custom Java-based Web Application Firewall (WAF); integrated mTLS, OAuth2, and OWASP Top 10 Mitigation strategies to neutralize injection and XSS vulnerabilities, reducing the security attack surface by 80%.
 - Redesigned healthcare prior authorization and appeals workflows by transforming IBM BPM/Camunda into Spring Boot microservices with Kafka-driven ETL, achieving 21% throughput gain and 18% cost reduction.
- Principal Software Engineer (Client: Fidelity Investments), Photon**
- Led migration of monolithic systems to microservices architecture, improving release cycles by 45%.
 - Designed Scalable API Platforms using Spring Boot and AWS API Gateway; optimized P99 latency for distributed systems and strengthened security by 80% through a custom Java-based WAF.
 - Implemented System Reliability by 35% via automated self-healing patterns in EKS; significantly reduced P1/P2 incidents by implementing Prometheus/Grafana observability stacks.

Previous Experience

Senior Technical Lead – Alten Calsoft (Jan 2019–Oct 2019)

Senior Associate – Cognizant (November 2015 – July 2018)

Lead Software Engineer – Collabera Technologies Pvt. Ltd (January 2014 – July 2014)

Lead Engineer – HCL Technologies (November 2010 – October 2012)

Senior Developer – Esquire Systems (November 2007 – September 2010)

Software Engineer – ACE Infotech (June 2004 – November 2007)

Education

MCA, Bharathidasan University (2004) | B.Sc Computer Science(2001)

Certifications

Cloud AWS Certified Solutions Architect – Associate & Professional (In Progress)

Java Sun Certified Java Professional – SCJP

Agile Certified Agile Scrum Master (ASM) – EXIN

AI/Python Python Programming Expert | Advanced GenAI (Upgrad) | Chatbot Dev (Alison)

Productivity AI Accelerator & Productivity Mastery – Be10X

Portfolio

<https://kayalshan.github.io/portfolio/>